Graph Theory

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Last updated: March 16, 2026

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1 Graphs

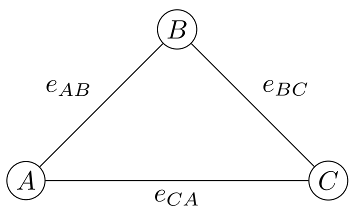
Definition 1.1 (Graphs)**.** A graph is made up of vertices and edges. An edge either connects two vertices or one vertex to itself.

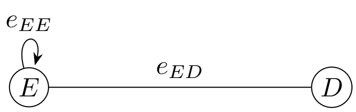
Remark. We will show 3 ways to represent data.

1. Simple list:

$$\begin{array}{ccccc} A & B & C & D & E \\ AB & AC & BC & DE & EE \end{array}$$

2. Graph form:





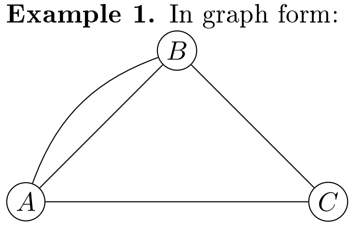
3. Matrix form:

	A	B	C	D	E
A	0	1	1	0	0
B	1	0	1	0	0
C	1	1	0	0	0
D	0	0	0	0	1
E	0	0	0	1	1

Definition 1.2 (Loop)**.** A loop is an edge which connects a vertex to itself.

Definition 1.3 (Parallel edges)**.** Parallel edges are two or more edges which connect the same vertices.

Example 1. In graph form:



Definition 1.4 (Simple graph)**.** A graph that contains no loops or parallel edges.

Definition 1.5 (Path)**.** A path is a sequence of vertices and edges that connects a starting vertex to an ending vertex without repeating any vertices.

Definition 1.6 (Trail)**.** A trail is a sequence of vertices and edges that connects a starting vertex to an ending vertex without repeating any edges.

Definition 1.7 (Walk)**.** A walk is a finite, alternating sequence of vertices and edges, which defines traversal through a graph.

Definition 1.8 (Circuit)**.** A circuit is a path that starts and ends at the same vertex, representing a closed loop.

Definition 1.9 (Hamiltonian Path)**.** A Hamiltonian path is a path that contains all of the graph vertices.

Definition 1.10 (Hamiltonian Circuit)**.** A Hamiltonian circuit is a circuit that contains all of the graph vertices.

Definition 1.11 (Euler Path)**.** An Euler path is a trail that contains all of the graph edges.

Definition 1.12 (Euler Circuit)**.** An Euler circuit is a circuit that contains all of the graph edges.

Definition 1.13 (Degree)**.** The degree of a vertex is the amount of edges connected to the vertex. (A loop adds 2 to the degree of a vertex)

Definition 1.14 (Connected Graph)**.** A connected graph is a graph in which there exists a path between any two given vertices.

Definition 1.15 (Component)**.** A component of a graph is a maximal subgraph in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the graph.